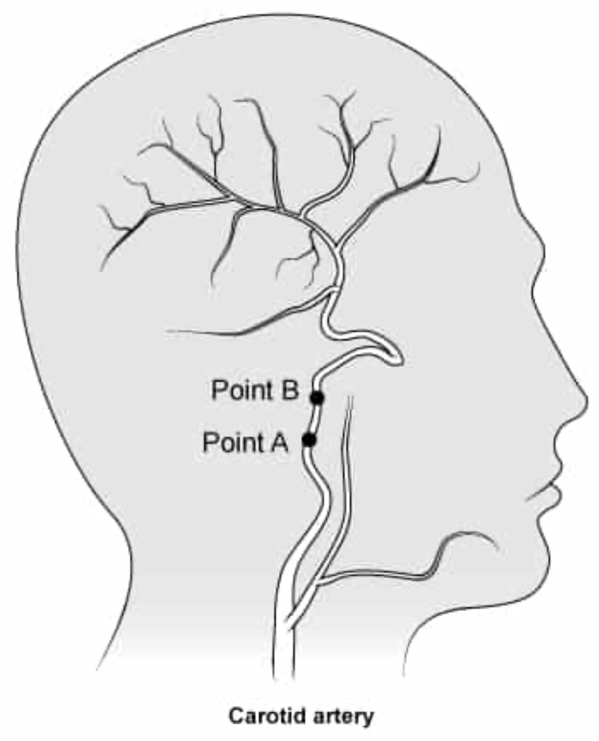
Blood flows 5 cm from Point A to Point B within an artery supplying the brain. The systolic blood pressure at Point A is 8 kPa.

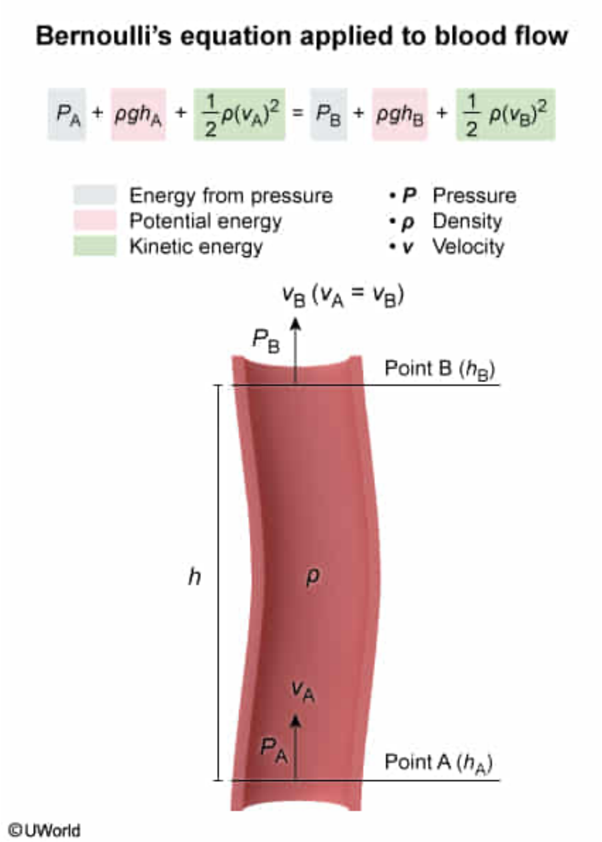


What is the systolic blood pressure when the blood reaches Point B? Assume the blood velocity and vessel diameter are constant over this distance. (Note: The density of blood is 1200 kg/m^3 and gravitational acceleration is 10 m/s^2.)

- A. 2.0 kPa [6%]
- ✔ B. 7.4 kPa [58%]
- C. 8.6 kPa [29%]
- D. 14 kPa [4%]

Correct
58% answered correctly

Explanation:



The total energy contained within a volume of fluid remains constant as it flows from one point to another within a conduit, following the law of conservation of energy. As a result, the total energy at any two points along the path of fluid flow must be equal:

$$(\text{Total Energy})_{\text{Point A}} = (\text{Total Energy})_{\text{Point B}}$$

Like solid objects, fluids possess both kinetic energy and potential energy between two points. The difference is that fluid kinetic and potential energy are both related to fluid **density (*ρ*)** rather than mass:

$$\rho\, g h_A + \frac{1}{2}\rho\,(v_A)^2 = \rho\, g h_B + \frac{1}{2}\rho\,(v_B)^2$$

Pressure (*P*) is also relevant to energy conservation in fluids because it may be expressed as a measurement of energy per unit volume. Adding terms for the energy contributions from pressure yields **Bernoulli's equation**, a mathematical expression for the conservation of potential energy, kinetic energy, and **pressure energy** of a fluid traveling through a conduit:

$$P_A + \rho\, g h_A + \frac{1}{2}\rho\,(v_A)^2 = P_B + \rho\, g h_B + \frac{1}{2}\rho\,(v_B)^2$$

In this question, blood has an initial systolic pressure P_A of 8 kPa at Point A. Because blood velocity and density are constant (v_A = v_B), the velocity terms can be eliminated from Bernoulli's equation and it can be rearranged to solve for the unknown pressure (P_B):

$$P_A + \rho g h_A + \cancel{\frac{1}{2}\rho (v_A)^2} = P_B + \rho g h_B + \cancel{\frac{1}{2}\rho (v_B)^2}$$

$$P_A + \rho g h_A = P_B + \rho g h_B$$

$$P_B = P_A + \rho g h_A - \rho g h_B$$

$$P_B = P_A + \rho g (h_A - h_B)$$

The blood travels 5 cm from Point A to Point B, so h_B is 5 cm greater than h_A, i.e. $h_A - h_B = -0.05$ m. Substituting this value and the initial pressure into the above equation yields:

$$P_B = 8000 \text{ Pa} + \left(1200 \text{ kg/m}^3\right) \cdot \left(10 \text{ m/s}^2\right) \cdot (-0.05 \text{ m})$$

$$P_B = 8000 \text{ kg/}\left(\text{m} \cdot \text{s}^2\right) - 600 \text{ kg/}\left(\text{m} \cdot \text{s}^2\right)$$

$$P_B = 7400 \text{ kg/}\left(\text{m} \cdot \text{s}^2\right) = 7.4 \text{ kPa}$$

Therefore, the pressure of blood that ascends 5 cm in the brain decreases from 8 kPa to 7.4 kPa when it reaches Point B.

(Choice A) 2.0 kPa is calculated if the distance of blood flow is converted incorrectly as −0.5 m.

(Choice C) 8.6 kPa is calculated if the distance of blood flow is incorrectly calculated as 0.05 m.

(Choice D) 14 kPa is calculated if the distance of blood flow is incorrectly calculated as 0.5 m.

Educational objective:
Fluid flows in accordance with the conservation of energy. Bernoulli's equation can be used to determine the total potential, kinetic, and pressure energy of a fluid at two points within a conduit.

Time spent:0 sec
Subject:
Physics

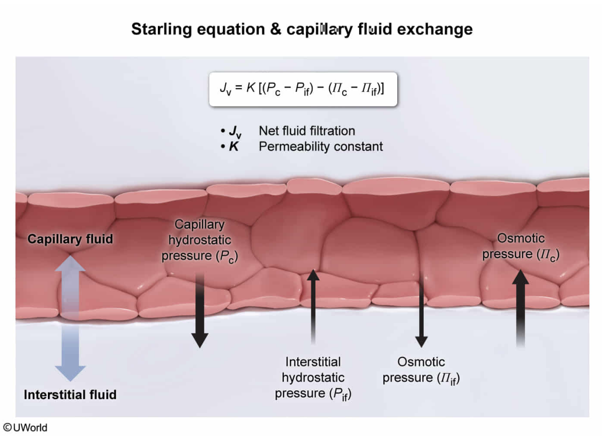
QID:402141
System:
Fluids

A scientist studying capillary fluid exchange in a laboratory environment can most effectively increase the net fluid filtration out of the capillaries by doing which of the following?

- A. Increasing the speed of blood flow [14%]
- B. Increasing interstitial hydrostatic pressure [39%]
- C. Replacing blood with a less viscous fluid [11%]
- ✔ D. Decreasing blood osmotic pressure [34%]

Correct
33% answered correctly

Explanation:



The movement of fluid into and out of capillaries is important in both physiology and disease. The **semipermeable membrane** separating capillaries from surrounding tissue (ie, the interstitial space) permits the movement of fluid between these compartments. Consequently, **net fluid filtration (J_v)** is determined by the hydrostatic and osmotic pressure within each compartment, where a (usually) positive J_v indicates net movement of fluid out of the capillary.

Hydrostatic pressure is created by fluid columns irrespective of spatial orientation (eg, blood vessels throughout the body). Fluid moves away from areas of high hydrostatic pressure and toward areas of low hydrostatic pressure. Consequently, capillary hydrostatic pressure (P_c) promotes the movement of fluid out of capillaries, whereas interstitial fluid hydrostatic pressure (P_{if}) diminishes the movement of fluid out of capillaries. Net fluid filtration is related to the difference in hydrostatic pressures as:

$$J_v \propto (P_c - P_{if})$$

Osmotic pressure (Π) is created during osmosis by the diffusion of solvent across a semipermeable membrane separating compartments with different solute concentrations. Fluid moves from areas of low osmotic pressure to areas of high osmotic pressure, which is relevant to blood vessel physiology because the osmotic pressure of blood is greater than the osmotic pressure of interstitial fluid.

The **Starling equation** can be used to calculate net fluid filtration and relates **membrane permeability (*K*)** to hydrostatic and osmotic pressure within the capillary and interstitial space:

$$J_{\mathrm{v}} = K\left[(P_{\mathrm{c}} - P_{\mathrm{if}}) - (\Pi_{c} - \Pi_{\mathrm{if}})\right]$$

Therefore, only a decrease in the osmotic pressure within the capillary Π_{c} (eg, reduced concentration of plasma proteins or salts) would result in a further net positive J_{v}, increasing the movement of fluid out of the capillary.

(Choices A and C) Increased blood speed or decreased blood viscosity would both result in greater volumetric blood flow. This would reduce blood hydrostatic pressure and *decrease* net fluid filtration (ie, less fluid moves out of the capillary).

(Choice B) Increasing the interstitial hydrostatic pressure inhibits the movement of fluid out of the capillaries and *decreases* net fluid filtration.

Educational objective:

Fluid compartments within the body have unique hydrostatic and osmotic pressures. Blood capillaries and interstitial fluid are separated by a semipermeable membrane, and the Starling equation may be used to determine net fluid filtration between these compartments.

Time spent: 0 sec | **QID:** 402142

Subject: Physics | **System:** Fluids